

PHYSICS 101 · ANSWER KEY - PROFESSOR COPY

Homework 1: Pressure

Grading guide. Total 21 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — What pressure is

1. In your own words, what is pressure? (Use the words push and area.) (2 pts)

Answer: Pressure is how much push lands on each bit of area — a push spread over an area. Same push on a smaller area = more pressure. 1 pt for 'push', 1 pt for 'spread over area / per area'.

2. You press down with the same push on a block that covers 1 square, and then on a block that covers 4 squares. Which block gets the higher pressure? How many times higher? (2 pts)

Answer: The 1-square block. 4 times higher (the push is spread over 4 times less area).

Part B — Match the unit to what it measures

3. Draw a line from each unit to the NICU measurement that uses it. (4 pts)

Answer: mmHg -> Blood pressure | cmH₂O -> CPAP pressure | percent (%) -> How much of the air is oxygen | grams (g) -> A baby's weight (1 pt each)

Part C — Torricelli's tube

4. Draw Torricelli's experiment: a dish of mercury with an upside-down tube standing in it, the mercury column inside labeled 760 mm. Next to it, draw a much shorter column and label it with a newborn's top blood pressure number (about 60 mmHg). (2 pts)

Answer: Dish + tube with a column labeled ~760 mm (1 pt); a clearly shorter column labeled about 60 mmHg (1 pt). Proportions don't need to be exact — shorter is the point.

Part D — Biggest push to smallest

5. Put these four pushes in order from BIGGEST to SMALLEST, then explain how you decided: (a) the atmosphere at sea level, 760 mmHg; (b) an adult's top blood pressure, 120 mmHg; (c) a newborn's top blood pressure, 70 mmHg; (d) CPAP at 6 cmH₂O (about 4 mmHg). (3 pts)

Answer: Atmosphere (760) > adult BP (120) > newborn BP (70) > CPAP (about 4 mmHg). 2 pts for the order; 1 pt for explaining that you converted CPAP to mmHg (or noticed cmH₂O is a much smaller unit) so everything could be compared in one unit.

6. Why do NICU teams use the gentlest ventilator pressure that still works? (1 pt)

Answer: A. Too much pressure overstretches delicate lungs and can pop an air sac. - Pressure can heal and pressure can hurt — 'gentle ventilation' protects lungs from scarring (BPD) and air leaks.

Part E — Flashback to NICU Tutor**7. A blood pressure is written “64/38.” What is each number called, and what does each one measure? (2 pts)**

Answer: 64 = systolic, the peak push during the heart's squeeze; 38 = diastolic, the low push between beats while the heart refills.

8. What does CPAP stand for, and what does it do for a preemie's lungs? (2 pts)

Answer: Continuous Positive Airway Pressure. A steady air cushion through nose prongs keeps the alveoli partly inflated so each breath starts from 'partly open' instead of collapsed.

Part F — Think like a doctor**9. A preemie on CPAP at 5 cmH₂O is still working hard to breathe — you can see the skin pulling in between the ribs. The team decides to raise the pressure a little, to 6 or 7. Why don't they just turn it way up, to 20, to be safe? (3 pts)**

Answer: More pressure does help hold the lungs open (1 pt), but too much overstretches and injures the fragile alveoli — scarring (BPD) or even popping one (air leak) — and can make it harder for blood to flow through the chest (1 pt). So the rule is the smallest pressure that works, raised in small steps while watching the baby (1 pt).